A MULTIVARIABLE CLASS OF COMPLETELY MONOTONIC FUNCTIONS

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ABSTRACT. Let $d \geq 1$, $c > 0$, and $\boldsymbol{b} \in \mathbb{R}^d$. For $\boldsymbol{x} \in (0,\infty)^d$, define

$$P_c^{(d)}(\boldsymbol{x}) := \sum_{\boldsymbol{m} \in \mathbb{N}_0^d} \frac{\boldsymbol{x}^{\boldsymbol{m}}}{\Gamma(c + |\boldsymbol{m}|)},$$

$$G_{\boldsymbol{b},c}^{(d)}(\boldsymbol{x}) := \left(\prod_{j=1}^{d} x_j^{-b_j} \right) \mathrm{e}^{-\sum_{j=1}^{d} x_j} P_c^{(d)}(\boldsymbol{x}).$$

We prove that $G_{\boldsymbol{b},c}^{(d)}$ is completely monotonic on $(0,\infty)^d$ if and only if $b_j \geq 0$ for every j and $c + \sum_{j \in J} b_j \geq |J|$ for every nonempty subset $J \subseteq \{1,\dots,d\}$. This extends a result of Alzer and Kwong, which is recovered when $d = 1$. We also establish a sharp functional inequality for $D_{c,d}(t) := P_c^{(d)}(t,\dots,t)$. For $0 < c < d$ and $x, y > 0$,

$$\mathrm{e} < \frac{D_{c,d}(1/x)^x D_{c,d}(1/y)^y}{D_{c,d}(1/(x+y))^{x+y}},$$

whereas the reverse strict inequality holds for $c > d$. In both cases, the constant e is best possible.

1. Introduction and main result

Throughout this paper, $\mathbb{N}_0 = \{0, 1, 2, \dots\}$ and $d \geq 1$ is an integer. For $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_d) \in \mathbb{N}_0^d$, we write

$$|\boldsymbol{\alpha}| = \sum_{j=1}^{d} \alpha_j, \qquad \boldsymbol{\alpha}! = \prod_{j=1}^{d} \alpha_j!, \qquad \partial^{\boldsymbol{\alpha}} = \frac{\partial^{|\boldsymbol{\alpha}|}}{\partial x_1^{\alpha_1} \cdots \partial x_d^{\alpha_d}}.$$

For $\boldsymbol{x}, \boldsymbol{t} \in \mathbb{R}^d$, we use $\boldsymbol{x}^{\boldsymbol{\alpha}} = \prod_{j=1}^{d} x_j^{\alpha_j}$ and $\langle \boldsymbol{x}, \boldsymbol{t} \rangle = \sum_{j=1}^{d} x_j t_j$, with the convention $x_j^0 = 1$. For a subset $J \subseteq \{1,\dots,d\}$, $|J|$ denotes its cardinality. Empty sums and products are understood to be 0 and 1, respectively. We denote Euler's gamma function by Γ and the support of a measure μ by $\operatorname{supp} \mu$.

A function $F \in C^\infty((0,\infty)^d)$ is *completely monotonic* (CM) if

$$(-1)^{|\boldsymbol{\alpha}|} \partial^{\boldsymbol{\alpha}} F(\boldsymbol{x}) \geq 0, \quad \boldsymbol{\alpha} \in \mathbb{N}_0^d,\ \boldsymbol{x} \in (0,\infty)^d.$$

By the multivariable Bernstein–Hausdorff–Widder theorem, this is equivalent to being the Laplace transform of a positive measure on $[0,\infty)^d$. Complete monotonic functions have a range of applications in classical analysis, special functions, probability, numerical analysis, moment problems, and operator theory; see, for example, [3, 7, 8].

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In [2], Alzer and Kwong considered the functions

$$g_{a,c}(x) := x^{-a}\mathrm{e}^{-x}S_c(x), \quad S_c(x) := \sum_{k=0}^{\infty} \frac{x^k}{\Gamma(c+k)}$$

where $c > 0$, and proved the following result [2, Theorem 3.1].

Theorem 1.1. *Let $a \in \mathbb{R}$ and $c > 0$. Then $g_{a,c} \in \mathrm{CM}((0,\infty)^1)$ if and only if $a \geq 0$, and $a + c \geq 1$.*

The purpose of this article is to present a complete multivariable extension of Theorem 1.1. A further motivation comes from the one-variable setting, where complete monotonicity provides short proofs of several functional inequalities.

Let $d \geq 1$, $c > 0$, $\boldsymbol{b} = (b_1, \ldots, b_d) \in \mathbb{R}^d$, and $\boldsymbol{x} = (x_1, \ldots, x_d) \in (0,\infty)^d$. Define

$$P_c^{(d)}(\boldsymbol{x}) := \sum_{\boldsymbol{m} \in \mathbb{N}_0^d} \frac{\boldsymbol{x}^{\boldsymbol{m}}}{\Gamma(c + |\boldsymbol{m}|)},$$

$$G_{\boldsymbol{b},c}^{(d)}(\boldsymbol{x}) := \left(\prod_{j=1}^{d} x_j^{-b_j} \right) \mathrm{e}^{-(x_1+\cdots+x_d)} P_c^{(d)}(\boldsymbol{x}).$$

The defining series converges locally uniformly in $\boldsymbol{x}$, together with all its partial derivatives, uniformly for c in compact subintervals of $(0,\infty)$; see Lemma 2.2 below. In particular, $G_{\boldsymbol{b},c}^{(d)}$ is smooth on $(0,\infty)^d$. Related classes appear in [1, 4, 5].

In several variables, each nonempty set of coordinates yields a necessary condition for the complete monotonicity of $G_{\boldsymbol{b},c}^{(d)}$; surprisingly, these conditions are also sufficient. The following is the main result of this paper.

Theorem 1.2 (Main theorem)**.** *Let $d \geq 1$, $c > 0$, and $\boldsymbol{b} = (b_1, \ldots, b_d) \in \mathbb{R}^d$. Then $G_{\boldsymbol{b},c}^{(d)} \in \mathrm{CM}((0,\infty)^d)$ if and only if*

$$b_j \geq 0, \qquad 1 \leq j \leq d, \tag{1.1}$$

and

$$c + \sum_{j \in J} b_j \geq |J| \qquad (\varnothing \neq J \subseteq \{1, \ldots, d\}). \tag{1.2}$$

For $d = 1$, this is the theorem of Alzer and Kwong. For $d = 2$, the conditions are

$$b_1, b_2 \geq 0, \quad b_1 + c \geq 1, \quad b_2 + c \geq 1, \quad \text{and} \quad b_1 + b_2 + c \geq 2.$$

The proof has two main ingredients. For sufficiency, we use a d-dimensional Dirichlet integral to express $G_{\boldsymbol{b},c}^{(d)}$ as a positive mixture of products of the one-variable functions from [2]; for necessity, we use partial diagonals and their growth at infinity.

2. Preliminaries

We recall the multivariable Bernstein–Hausdorff–Widder theorem; see [3, Proposition 4.6.11].

Proposition 2.1. *A function $F : (0,\infty)^d \to \mathbb{R}$ belongs to $\mathrm{CM}((0,\infty)^d)$ if and only if there is a positive Radon measure μ on $[0,\infty)^d$ such that*

$$F(\boldsymbol{x}) = \int_{[0,\infty)^d} \mathrm{e}^{-\langle \boldsymbol{x},\boldsymbol{t}\rangle}\, d\mu(\boldsymbol{t}).$$

The integral is required to be finite for every $\boldsymbol{x} \in (0,\infty)^d$.

We first record the convergence properties of the defining series.

Lemma 2.2. *The series defining $P_\gamma^{(d)}(\boldsymbol{x})$, together with all its termwise partial derivatives with respect to $\boldsymbol{x}$, converges uniformly on $K \times I$ for every compact set $K \subset \mathbb{R}^d$ and every compact interval $I \subset (0,\infty)$. Consequently, $P_\gamma^{(d)}$ is C^∞ in $\boldsymbol{x}$, and $(\boldsymbol{x},\gamma) \mapsto \partial^{\boldsymbol{\alpha}} P_\gamma^{(d)}(\boldsymbol{x})$ is jointly continuous on $\mathbb{R}^d \times (0,\infty)$ for every multi-index $\boldsymbol{\alpha}$.*

Proof. Fix K and $I = [a, A] \subset (0,\infty)$, and choose $R \geq 1$ such that $|x_j| \leq R$ on K for every j. Set $C_I = \min_{\gamma\in I} \Gamma(\gamma) > 0$. For $k \geq 1$,

$$\Gamma(\gamma + k) = \Gamma(\gamma)\gamma \prod_{\ell=1}^{k-1} (\gamma + \ell) \geq C_I a (k-1)!, \qquad \gamma \in I.$$

Let $\boldsymbol{\alpha} \in \mathbb{N}_0^d$ and $N = |\boldsymbol{\alpha}|$. If $|\boldsymbol{m}| = k \geq 1$, then

$$\sup_{\boldsymbol{x}\in K} |\partial^{\boldsymbol{\alpha}} \boldsymbol{x}^{\boldsymbol{m}}| \leq k^N R^k,$$

with the derivative understood to be zero whenever $m_j < \alpha_j$ for some j. Since there are $\binom{k+d-1}{d-1}$ multi-indices of total degree k, the terms of positive degree are uniformly majorized on $K \times I$ by

$$\frac{1}{C_I a} \sum_{k=1}^{\infty} \binom{k+d-1}{d-1} \frac{k^N R^k}{(k-1)!},$$

which converges by the ratio test. The degree-zero term is either $1/\Gamma(\gamma)$ or zero. The Weierstrass M-test therefore yields the asserted uniform convergence. Local uniform convergence of the series and all its differentiated series justifies termwise differentiation, while continuity of the summands gives joint continuity of every resulting partial derivative. □

The next lemma reduces the family of subset inequalities to one inequality.

Lemma 2.3. *Let $b_1, \ldots, b_d \geq 0$ and $c > 0$. The inequalities*

$$c + \sum_{j\in J} b_j \geq |J| \qquad (\varnothing \neq J \subseteq \{1,\ldots,d\}) \tag{2.1}$$

are equivalent to

$$c \geq \sum_{j=1}^{d} \max\{0, 1 - b_j\}. \tag{2.2}$$

Proof. Condition (2.1) yields that $c \geq \sum_{j\in J}(1 - b_j)$ for every nonempty J. If some $b_j < 1$, the largest such sum is obtained by taking precisely the indices with $b_j < 1$, and equals the right-hand side of (2.2). If every $b_j \geq 1$, both conditions are automatic since $c > 0$. □

The next elementary fact explains why partial diagonals appear in the proof of necessity.

Lemma 2.4. *Let $F \in \mathrm{CM}((0,\infty)^d)$. Let $\boldsymbol{a} \in [0,\infty)^d \setminus \{\mathbf{0}\}$ and let $\boldsymbol{z} \in [0,\infty)^d$ be such that $\boldsymbol{z} + t\boldsymbol{a} \in (0,\infty)^d$ for every $t > 0$. Then*

$$t \longmapsto F(\boldsymbol{z} + t\boldsymbol{a})$$

belongs to $\mathrm{CM}((0,\infty)^1)$.

Proof. For $N \in \mathbb{N}_0$, the chain rule gives

$$\frac{d^N}{dt^N} F(\boldsymbol{z} + t\boldsymbol{a}) = \sum_{|\boldsymbol{\alpha}|=N} \frac{N!}{\boldsymbol{\alpha}!} \boldsymbol{a}^{\boldsymbol{\alpha}} \partial^{\boldsymbol{\alpha}} F(\boldsymbol{z} + t\boldsymbol{a}).$$

Multiplication by $(-1)^N$ and the nonnegativity of all coefficients prove the assertion. □

We write $f(x) \sim g(x)$ as $x \to a$ when $\lim_{x\to a} f(x)/g(x) = 1$. Recall that $S_a(t) = \sum_{k=0}^{\infty} t^k/\Gamma(a+k)$ for $a > 0$. We now record the diagonal asymptotics of the multivariable series.

Lemma 2.5. *Let $r \geq 1$ and $a > 0$. Then*

$$P_a^{(r)}(t,\dots,t) \sim \frac{1}{(r-1)!} \mathrm{e}^t t^{r-a} \qquad (t \to \infty).$$

Proof. Grouping the series according to the total degree gives

$$P_a^{(r)}(t,\dots,t) = \sum_{k=0}^{\infty} \binom{k+r-1}{r-1} \frac{t^k}{\Gamma(a+k)}.$$

Let $\vartheta = t\frac{d}{dt}$. Since $\vartheta(t^k) = kt^k$,

$$\frac{1}{(r-1)!} \prod_{\ell=1}^{r-1} (\vartheta + \ell) t^k = \binom{k+r-1}{r-1} t^k.$$

and hence

$$P_a^{(r)}(t,\dots,t) = \frac{1}{(r-1)!} \prod_{\ell=1}^{r-1} (\vartheta + \ell) S_a(t). \tag{2.3}$$

For $a > 0$, the beta integral and termwise integration give

$$\begin{aligned} S_a(t) &= \frac{1}{\Gamma(a)} + \frac{t}{\Gamma(a)} \int_0^1 u^{a-1} e^{t(1-u)}\, du \\ &= \frac{1}{\Gamma(a)} + \frac{e^t t^{1-a}}{\Gamma(a)} \int_0^t v^{a-1} e^{-v}\, dv. \end{aligned}$$

Indeed, expanding the exponential in the first integral gives $t^{k+1}/\Gamma(a+k+1)$ for its kth integrated term. Since the last integral tends to $\Gamma(a)$, the one-variable series satisfies

$$S_a(t) \sim \mathrm{e}^t t^{1-a} \qquad (t \to \infty) \tag{2.4}$$

and

$$\vartheta S_a(t) = (t + 1 - a) S_a(t) - \frac{1-a}{\Gamma(a)}. \tag{2.5}$$

An induction based on (2.5) gives

$$\prod_{\ell=1}^{N}(\vartheta+\ell)S_a(t) = Q_N(t)S_a(t) + R_N(t),$$

where Q_N is a monic polynomial of degree N and R_N is a polynomial. Indeed, starting with $Q_0 = 1$ and $R_0 = 0$, one obtains

$$\begin{aligned} Q_{N+1}(t) &= tQ_N'(t) + (t+N+2-a)Q_N(t),\\ R_{N+1}(t) &= -\frac{1-a}{\Gamma(a)}Q_N(t) + tR_N'(t) + (N+1)R_N(t). \end{aligned}$$

Equation (2.4) now yields

$$\prod_{\ell=1}^{N}(\vartheta+\ell)S_a(t) \sim \mathrm{e}^t t^{N+1-a}.$$

This together with (2.3) proves the result. □

We will also use the following elementary consequence of the Bernstein–Widder theorem.

Lemma 2.6. *Let $0 \neq h \in \mathrm{CM}((0,\infty)^1)$, and let μ be its representing measure. If $a := \inf\operatorname{supp}\mu$, then*

$$\lim_{x\to\infty} -\frac{1}{x}\log h(x) = a. \tag{2.6}$$

Moreover, the function $x \longmapsto \mathrm{e}^{ax}h(x)$ is completely monotonic and, in particular, nonincreasing.

Proof. Fix $x_0 > 0$. For every $\varepsilon > 0$, the definition of a and the local finiteness of μ give

$$0 < c_\varepsilon := \mu([a, a+\varepsilon]) < \infty.$$

Since μ is supported on $[a,\infty)$, for $x \geq x_0$,

$$c_\varepsilon \mathrm{e}^{-x(a+\varepsilon)} \leq h(x) \leq \mathrm{e}^{-(x-x_0)a}h(x_0).$$

Taking logarithms, dividing by x, and letting $x \to \infty$, we obtain

$$-a-\varepsilon \leq \liminf_{x\to\infty}\frac{\log h(x)}{x} \leq \limsup_{x\to\infty}\frac{\log h(x)}{x} \leq -a.$$

Letting $\varepsilon \downarrow 0$ proves (2.6).

Now translate μ by a. More precisely, define a positive measure $\widetilde{\mu}$ on $[0,\infty)$ by $\widetilde{\mu}(E) = \mu(E+a)$. Then

$$\mathrm{e}^{ax}h(x) = \int_{[0,\infty)} \mathrm{e}^{-xu}\, d\widetilde{\mu}(u).$$

Hence $x \mapsto \mathrm{e}^{ax}h(x)$ is CM and therefore nonincreasing. □

Lemma 2.7. *Fix $\boldsymbol{b} \in \mathbb{R}^d$ and $c > 0$, and let $c_n > 0$ with $c_n \to c$. If $G^{(d)}_{\boldsymbol{b},c_n} \in \mathrm{CM}((0,\infty)^d)$ for every n, then $G^{(d)}_{\boldsymbol{b},c} \in \mathrm{CM}((0,\infty)^d)$.*

Proof. Since $\mathrm{CM}((0,\infty)^d)$ is closed under pointwise limits, it suffices to show that

$$P_{c_n}^{(d)}(\boldsymbol{x}) \longrightarrow P_c^{(d)}(\boldsymbol{x}), \qquad \boldsymbol{x} \in (0,\infty)^d.$$

Choose a compact interval $I \subset (0,\infty)$ containing c and all sufficiently large c_n. By Lemma 2.2, $P_\gamma^{(d)}(\boldsymbol{x})$ is continuous in $\gamma \in I$, which gives the required convergence. The prefactor in $G_{\boldsymbol{b},c}^{(d)}$ is finite and independent of c. □

3. Proof of the main theorem

In this section we give a proof of Theorem 1.2. To do this, we separate necessity from sufficiency.

Proposition 3.1 (Necessary conditions)**.** *If* $G_{\boldsymbol{b},c}^{(d)} \in \mathrm{CM}((0,\infty)^d)$*, then* (1.1) *and* (1.2) *hold.*

Proof. Fix $j \in \{1,\dots,d\}$ and all coordinates other than x_j. Then

$$h_j(x_j) := G_{\boldsymbol{b},c}^{(d)}(\boldsymbol{x})$$

is completely monotonic and hence nonincreasing. By Lemma 2.2, as $x_j \downarrow 0$, the series $P_c^{(d)}(\boldsymbol{x})$ converges to a finite limit, which is positive because its constant term is $1/\Gamma(c) > 0$. Hence $h_j(x_j) \sim C_j x_j^{-b_j}$ for some $C_j > 0$. If $b_j < 0$, then $h_j(x_j) \to 0$, which contradicts positivity and monotonicity. Thus $b_j \geq 0$.

Let J be a nonempty subset of $\{1,\dots,d\}$ and put $r = |J|$. Fix $z_k > 0$ for $k \notin J$, and define $\boldsymbol{x}(t)$ by

$$x_j(t) = t \quad (j \in J), \qquad x_k(t) = z_k \quad (k \notin J).$$

Set

$$h_J(t) := G_{\boldsymbol{b},c}^{(d)}(\boldsymbol{x}(t)).$$

By Lemma 2.4, $h_J \in \mathrm{CM}((0,\infty)^1)$.

Retaining only the positive terms with $m_k = 0$ for $k \notin J$ gives

$$P_c^{(d)}(\boldsymbol{x}(t)) \geq \sum_{\substack{\boldsymbol{m}\in\mathbb{N}_0^d \\ m_k=0 \text{ for all } k\notin J}} \frac{t^{\sum_{j\in J} m_j}}{\Gamma\Big(c+\sum_{j\in J} m_j\Big)} = P_c^{(r)}(t,\dots,t), \qquad r = |J|.$$

By Lemma 2.5,

$$P_c^{(r)}(t,\dots,t) \sim \frac{\mathrm{e}^t t^{r-c}}{(r-1)!}.$$

Consequently, for all sufficiently large t,

$$P_c^{(d)}(\boldsymbol{x}(t)) \geq P_c^{(r)}(t,\dots,t) \geq C_1 \mathrm{e}^t t^{r-c}, \tag{3.1}$$

where $C_1 = 1/[2(r-1)!] > 0$.

For large t, each $z_k \leq t$; hence, for every $\boldsymbol{m} \in \mathbb{N}_0^d$,

$$\prod_{j=1}^{d} x_j(t)^{m_j} \leq t^{|\boldsymbol{m}|}.$$

Termwise comparison therefore yields

$$P_c^{(d)}(\boldsymbol{x}(t)) \le \sum_{\boldsymbol{m}\in\mathbb{N}_0^d} \frac{t^{|\boldsymbol{m}|}}{\Gamma(c+|\boldsymbol{m}|)} = P_c^{(d)}(t,\dots,t)$$

for sufficiently large t. Lemma 2.5, applied to the lower and upper bounds, now gives

$$\lim_{t\to\infty} \frac{1}{t} \log P_c^{(d)}(\boldsymbol{x}(t)) = 1. \tag{3.2}$$

Put $B_J = \sum_{j\in J} b_j$. Since the fixed complementary coordinates contribute a positive constant,

$$h_J(t) = C_2 t^{-B_J} \mathrm{e}^{-rt} P_c^{(d)}(\boldsymbol{x}(t)).$$

Together with (3.2), this yields

$$\lim_{t\to\infty} -\frac{1}{t} \log h_J(t) = r - 1.$$

By Lemma 2.6, $\mathrm{e}^{(r-1)t} h_J(t)$ is nonincreasing and hence bounded for large t, whereas (3.1) gives

$$\mathrm{e}^{(r-1)t} h_J(t) \ge C_3 t^{r-c-B_J}.$$

Thus $r - c - B_J \le 0$, so $c + \sum_{j\in J} b_j \ge r = |J|$. Since J was arbitrary, (1.2) follows. □

We next prove sufficiency when the reduced parameter inequality is strict.

Proposition 3.2. *Suppose that $b_j \ge 0$ for $1 \le j \le d$ and*

$$c > \sum_{j=1}^{d} \max\{0, 1 - b_j\}.$$

Then $G_{\boldsymbol{b},c}^{(d)} \in \mathrm{CM}((0,\infty)^d)$.

Proof. Put $A_j = \max\{0, 1-b_j\}$ and $\delta = c - \sum_{j=1}^d A_j > 0$. Choose

$$p_j = A_j + \frac{\delta}{d+1}, \qquad q = \frac{\delta}{d+1}.$$

Then $p_1, \dots, p_d, q > 0$, $p_1 + \cdots + p_d + q = c$, and $p_j \ge A_j$. Thus

$$b_j + p_j \ge 1, \qquad 1 \le j \le d. \tag{3.3}$$

Let $\Delta_d := \left\{ \boldsymbol{s} \in (0,\infty)^d : \sum_{j=1}^d s_j < 1 \right\}$. By the Dirichlet integral [6, Eq. (5.14.2)], we obtain

$$\int_{\Delta_d} \left(\prod_{j=1}^{d} s_j^{m_j + p_j - 1} \right) \left(1 - \sum_{j=1}^{d} s_j \right)^{q-1} d\boldsymbol{s} = \frac{\Gamma(q) \prod_{j=1}^d \Gamma(m_j + p_j)}{\Gamma(c + |\boldsymbol{m}|)}. \tag{3.4}$$

Using (3.4) in the defining series and applying Tonelli's theorem, we get

$$P_c^{(d)}(\boldsymbol{x}) = \frac{1}{\Gamma(q)} \int_{\Delta_d} \left(1 - \sum_{j=1}^{d} s_j \right)^{q-1} \prod_{j=1}^{d} \left[s_j^{p_j - 1} S_{p_j}(x_j s_j) \right] d\boldsymbol{s}. \tag{3.5}$$

For $0 < s < 1$, define $h_j(x; s) := s^{b_j} \mathrm{e}^{-(1-s)x} (sx)^{-b_j} \mathrm{e}^{-sx} S_{p_j}(sx)$. Then

$$h_j(x; s) = x^{-b_j} \mathrm{e}^{-x} S_{p_j}(sx). \tag{3.6}$$

By Theorem 1.1 and (3.3), $x \longmapsto (sx)^{-b_j}\mathrm{e}^{-sx}S_{p_j}(sx)$ is CM. Hence $h_j(\,\cdot\,;s)$ is CM.

Multiplying (3.5) by $\prod_{j=1}^{d} x_j^{-b_j}\mathrm{e}^{-x_j}$ and using (3.6), we obtain

$$G_{\boldsymbol{b},c}^{(d)}(\boldsymbol{x}) = \frac{1}{\Gamma(q)}\int_{\Delta_d}\left(1-\sum_{j=1}^{d}s_j\right)^{q-1}\prod_{j=1}^{d}\left[s_j^{p_j-1}h_j(x_j;s_j)\right]d\boldsymbol{s}.$$

Since all derivatives of $h_j(x;s)$ in x are locally bounded uniformly for $s\in[0,1]$, and the Dirichlet weight is integrable, differentiation under the integral sign is justified by dominated convergence. The complete monotonicity of each $h_j(\,\cdot\,;s_j)$ therefore yields that for every $\boldsymbol{\alpha}\in\mathbb{N}_0^d$,

$$\begin{aligned}&(-1)^{|\boldsymbol{\alpha}|}\partial^{\boldsymbol{\alpha}}G_{\boldsymbol{b},c}^{(d)}(\boldsymbol{x})\\ &\quad=\frac{1}{\Gamma(q)}\int_{\Delta_d}\left(1-\sum_{j=1}^{d}s_j\right)^{q-1}\prod_{j=1}^{d}\left[s_j^{p_j-1}(-1)^{\alpha_j}\partial_{x_j}^{\alpha_j}h_j(x_j;s_j)\right]d\boldsymbol{s}\ge 0.\end{aligned}$$

This completes the proof. □

Proof of Theorem 1.2. The necessity follows from Proposition 3.1.

Conversely, suppose that (1.1) and (1.2) hold. By Lemma 2.3, $c\ge\sum_{j=1}^{d}\max\{0,1-b_j\}$. If the inequality is strict, Proposition 3.2 gives the result.

Suppose equality holds and set $c_n=c+1/n$. Then $c_n>\sum_{j=1}^{d}\max\{0,1-b_j\}$. Proposition 3.2 shows that $G_{\boldsymbol{b},c_n}^{(d)}\in\mathrm{CM}((0,\infty)^d)$. Since $c_n\to c>0$, Lemma 2.7 gives the result at c. □

4. Applications

In this section, we discuss more classes of completely monotonic functions arising as a consequence of our main theorem.

Alzer and Kwong [2, Corollary 3.2] proved that

$$H_p(x) := \frac{\mathrm{e}^{-x^p}}{x}\left(\mathrm{e}^{x^p}-1-x^p\right),\qquad x>0,$$

is completely monotonic if and only if $0<p\le 1/2$. Since

$$P_3^{(1)}(t)=\frac{\mathrm{e}^t-1-t}{t^2},$$

their function can be written as

$$H_p(x)=x^{2p-1}\mathrm{e}^{-x^p}P_3^{(1)}(x^p).$$

This representation motivates the following multivariable extension.

We first record the composition property needed below.

Lemma 4.1. *Let $F\in\mathrm{CM}((0,\infty)^d)$ and let $0<p_j\le 1$ for $j=1,\dots,d$. Then the function*

$$\boldsymbol{x}\longmapsto F(x_1^{p_1},\dots,x_d^{p_d})$$

belongs to $\mathrm{CM}((0,\infty)^d)$.

Proof. Put $\phi_j(x) = x^{p_j}$. For every integer $\ell \geq 1$,

$$(-1)^{\ell-1}\phi_j^{(\ell)}(x) \geq 0, \qquad x > 0.$$

Since ϕ is a Bernstein function for $0 < p_j \leq 1$, the coordinatewise composition property implies that $F(x_1^{p_1}, \ldots, x_d^{p_d})$ is completely monotonic. □

Theorem 4.2. *Let $d \geq 1$, $c \geq d$, and $\boldsymbol{p} = (p_1, \ldots, p_d) \in (0,\infty)^d$. Define*

$$H_{\boldsymbol{p},c}^{(d)}(\boldsymbol{x}) := \left(\prod_{j=1}^{d} x_j^{2p_j-1}\right) \mathrm{e}^{-\sum_{j=1}^{d} x_j^{p_j}} P_c^{(d)}(x_1^{p_1}, \ldots, x_d^{p_d}), \qquad \boldsymbol{x} \in (0,\infty)^d.$$

Then

$$H_{\boldsymbol{p},c}^{(d)} \in \mathrm{CM}((0,\infty)^d) \quad \Longleftrightarrow \quad 0 < p_j \leq \frac{1}{2} \quad \textit{for every } j = 1, \ldots, d.$$

Proof. Suppose first that $0 < p_j \leq 1/2$ for every j, and put

$$b_j := \frac{1}{p_j} - 2.$$

Then $b_j \geq 0$. Moreover, for every nonempty subset $J \subseteq \{1, \ldots, d\}$,

$$c + \sum_{j \in J} b_j \geq c \geq d \geq |J|.$$

Theorem 1.2 therefore yields $G_{\boldsymbol{b},c}^{(d)} \in \mathrm{CM}((0,\infty)^d)$. Since $-p_j b_j = 2p_j - 1$, we have

$$H_{\boldsymbol{p},c}^{(d)}(\boldsymbol{x}) = G_{\boldsymbol{b},c}^{(d)}(x_1^{p_1}, \ldots, x_d^{p_d}).$$

Lemma 4.1 now proves complete monotonicity.

Conversely, suppose that $H_{\boldsymbol{p},c}^{(d)} \in \mathrm{CM}((0,\infty)^d)$. Fix j and positive numbers z_k for $k \neq j$, and let $\boldsymbol{x}(t)$ have jth coordinate t and kth coordinate z_k for $k \neq j$. The function

$$h_j(t) := H_{\boldsymbol{p},c}^{(d)}(\boldsymbol{x}(t))$$

is positive and nonincreasing on $(0,\infty)$.

Local uniform convergence of the series defining $P_c^{(d)}$ gives

$$P_c^{(d)}(z_1^{p_1}, \ldots, t^{p_j}, \ldots, z_d^{p_d}) \longrightarrow L_j \qquad (t \downarrow 0),$$

where

$$L_j := \sum_{\substack{\boldsymbol{m} \in \mathbb{N}_0^d \\ m_j = 0}} \frac{\prod_{k \neq j} z_k^{p_k m_k}}{\Gamma(c + |\boldsymbol{m}|)}$$

is finite and satisfies $L_j \geq 1/\Gamma(c) > 0$. It follows that

$$h_j(t) \sim C_j t^{2p_j - 1} \qquad (t \downarrow 0),$$

with

$$C_j = L_j \left(\prod_{k \neq j} z_k^{2p_k - 1}\right) \mathrm{e}^{-\sum_{k \neq j} z_k^{p_k}} > 0.$$

If $p_j > 1/2$, then $h_j(t) \to 0$ as $t \downarrow 0$. However, for any fixed $t_0 > 0$, monotonicity gives

$$h_j(t) \geq h_j(t_0) > 0, \qquad 0 < t < t_0,$$

a contradiction. Hence $p_j \le 1/2$. Since j was arbitrary, the necessity follows. □

In particular, choosing $c = d+2$ and $p_1 = \cdots = p_d = p > 0$ gives the one-parameter family

$$H_p^{(d)}(\boldsymbol{x}) := \left(\prod_{j=1}^{d} x_j^{2p-1}\right) \mathrm{e}^{-\sum_{j=1}^{d} x_j^p} P_{d+2}^{(d)}(x_1^p, \dots, x_d^p).$$

Theorem 4.2 shows that

$$H_p^{(d)} \in \mathrm{CM}((0,\infty)^d) \quad \Longleftrightarrow \quad 0 < p \le \frac{1}{2}.$$

For $d = 1$, this function is exactly

$$H_p^{(1)}(x) = x^{2p-1}\mathrm{e}^{-x^p}\frac{\mathrm{e}^{x^p} - 1 - x^p}{x^{2p}} = \frac{\mathrm{e}^{-x^p}}{x}\left(\mathrm{e}^{x^p} - 1 - x^p\right),$$

so the original Alzer–Kwong corollary is recovered.

Positive integral mixtures yield further classes of completely monotonic functions. The following result extends [2, Theorem 3.3] to several variables, with coefficients depending on the total degree $|\boldsymbol{m}|$.

Theorem 4.3. *Let $d \ge 1$, $-1 < \alpha < \beta$, and let $\phi : (\alpha, \beta) \to [0, \infty)$ be measurable with*

$$\int_\alpha^\beta \frac{\phi(t)}{(t+1)\cdots(t+d)}\, dt < \infty.$$

For $\boldsymbol{m} \in \mathbb{N}_0^d$, define

$$a_{\boldsymbol{m}} := \int_\alpha^\beta \frac{\phi(t)}{\prod_{\ell=1}^{|\boldsymbol{m}|+d}(t+\ell)}\, dt.$$

Then the series

$$S_\phi^{(d)}(\boldsymbol{x}) := \mathrm{e}^{-\sum_{j=1}^{d} x_j} \sum_{\boldsymbol{m} \in \mathbb{N}_0^d} a_{\boldsymbol{m}} \boldsymbol{x}^{\boldsymbol{m}}$$

is finite for every $\boldsymbol{x} \in (0,\infty)^d$ and defines a completely monotonic function.

Proof. Set

$$\Delta_d = \left\{ \boldsymbol{s} \in (0,\infty)^d : \sum_{j=1}^{d} s_j < 1 \right\}, \qquad K_\phi(\boldsymbol{s}) = \int_\alpha^\beta \phi(t) \left(1 - \sum_{j=1}^{d} s_j\right)^t dt.$$

The Dirichlet integral gives

$$\int_{\Delta_d} \boldsymbol{s}^{\boldsymbol{m}} \left(1 - \sum_{j=1}^{d} s_j\right)^t d\boldsymbol{s} = \frac{\boldsymbol{m}!\,\Gamma(t+1)}{\Gamma(t+d+1+|\boldsymbol{m}|)}.$$

By Tonelli's theorem and the Dirichlet integral,

$$\int_{\Delta_d} \boldsymbol{s}^{\boldsymbol{m}} K_\phi(\boldsymbol{s})\, d\boldsymbol{s} = \boldsymbol{m}! \int_\alpha^\beta \frac{\phi(t)\Gamma(t+1)}{\Gamma(t+d+1+|\boldsymbol{m}|)}\, dt = \boldsymbol{m}!\, a_{\boldsymbol{m}}.$$

Expanding the exponential and applying Tonelli's theorem again gives

$$\int_{\Delta_d} e^{-\sum_{j=1}^d x_j(1-s_j)} K_\phi(\boldsymbol{s})\, d\boldsymbol{s} = e^{-\sum_{j=1}^d x_j} \sum_{\boldsymbol{m}\in\mathbb{N}_0^d} \frac{\boldsymbol{x}^{\boldsymbol{m}}}{\boldsymbol{m}!} \int_{\Delta_d} \boldsymbol{s}^{\boldsymbol{m}} K_\phi(\boldsymbol{s})\, d\boldsymbol{s}$$
$$= S_\phi^{(d)}(\boldsymbol{x}).$$

Since $\int_{\Delta_d} K_\phi(\boldsymbol{s})\, d\boldsymbol{s} = a_{\boldsymbol{0}} < \infty$ and $0 < 1 - s_j < 1$, differentiation under the integral is justified by dominated convergence. Thus, for every $\boldsymbol{\nu} \in \mathbb{N}_0^d$,

$$(-1)^{|\boldsymbol{\nu}|} \partial^{\boldsymbol{\nu}} S_\phi^{(d)}(\boldsymbol{x}) = \int_{\Delta_d} \prod_{j=1}^d (1-s_j)^{\nu_j} e^{-\sum_{j=1}^d x_j(1-s_j)} K_\phi(\boldsymbol{s})\, d\boldsymbol{s} \geq 0,$$

which proves complete monotonicity. □

5. A sharp functional inequality on the diagonal

In this section, we establish a diagonal counterpart of the sharp inequality of Alzer and Kwong [2, Theorem 4.3]. For $d \geq 1$ and $c > 0$, put

$$D_{c,d}(t) := P_c^{(d)}(t, \ldots, t), \qquad t \geq 0.$$

We first determine the logarithmic convexity of this function.

Lemma 5.1. *Let $d \geq 1$ and $c > 0$. The function $D_{c,d}$ is strictly log-concave on $(0,\infty)$ if $0 < c < d$, and strictly log-convex on $(0,\infty)$ if $c > d$. If $c = d$, then*

$$D_{d,d}(t) = \frac{e^t}{\Gamma(d)}, \qquad t \geq 0.$$

Proof. Note that

$$D_{c,d}(t) = \sum_{k=0}^\infty a_k t^k, \qquad a_k := \frac{\binom{k+d-1}{d-1}}{\Gamma(c+k)} > 0,\ t \geq 0.$$

The coefficients satisfy

$$r_k := \frac{(k+1)a_{k+1}}{a_k} = \frac{k+d}{k+c}.$$

Consequently,

$$D'_{c,d}(t) = \sum_{k=0}^\infty r_k a_k t^k, \qquad t \geq 0.$$

For brevity, write $D = D_{c,d}$. Since

$$D'(t) = \sum_{k=0}^\infty r_k a_k t^k = \sum_{k=1}^\infty k a_k t^{k-1}, \qquad t \geq 0,$$

absolute convergence allows us to expand and regroup:

$$\begin{aligned} D''(t)D(t) - (D'(t))^2 &= \sum_{\substack{i,j\geq 0\\ i+j\geq 1}} j(r_j - r_i)a_i a_j t^{i+j-1} \\ &= \sum_{0\leq i<j} \big[j(r_j - r_i) + i(r_i - r_j)\big] a_i a_j t^{i+j-1} \\ &= \sum_{0\leq i<j} (j-i)(r_j - r_i)a_i a_j t^{i+j-1} \\ &= (c-d) \sum_{0\leq i<j} \frac{(j-i)^2 a_i a_j}{(i+c)(j+c)} t^{i+j-1}. \end{aligned}$$

Here the diagonal terms vanish, and the second equality pairs the terms indexed by (i,j) and (j,i). In the last equality we used

$$r_j - r_i = \frac{(c-d)(j-i)}{(j+c)(i+c)}.$$

The last series is strictly positive for $t>0$, since all its terms are nonnegative and the term $(i,j)=(0,1)$ is positive. Therefore

$$\frac{d^2}{dt^2} \log D_{c,d}(t) \begin{cases} <0, & 0<c<d, \\ >0, & c>d. \end{cases}$$

Finally, if $c=d$, then

$$a_k = \frac{1}{\Gamma(d)k!},$$

which gives the asserted exponential formula. □

The following is the diagonal counterpart of the sharp inequality of [2, Theorem 4.3].

Theorem 5.2. *Let $d\geq 1$ and $c>0$. For all $x,y>0$, if $0<c<d$, then*

$$e < \frac{\big[P_c^{(d)}(1/x,\dots,1/x)\big]^x \big[P_c^{(d)}(1/y,\dots,1/y)\big]^y}{\big[P_c^{(d)}(1/(x+y),\dots,1/(x+y))\big]^{x+y}}. \tag{5.1}$$

If $c>d$, the reverse strict inequality holds in (5.1). *If $c=d$, the quotient equals e for every $x,y>0$. For each fixed d and $c\neq d$, the constant e is best possible.*

Proof. Define

$$J_{c,d}(x) := x \log D_{c,d}(1/x), \qquad x>0.$$

By Lemma 2.5,

$$D_{c,d}(t) \sim \frac{e^t t^{d-c}}{\Gamma(d)} \qquad (t\to\infty).$$

It follows that

$$J_{c,d}(x) = 1 - (d-c)x\log x - x\log\Gamma(d) + o(x) \qquad (x\downarrow 0).$$

In particular, $J_{c,d}$ extends continuously to $[0,\infty)$ by setting $J_{c,d}(0)=1$.

A direct differentiation gives

$$J''_{c,d}(x) = \frac{1}{x^3} \left.\frac{d^2}{dt^2} \log D_{c,d}(t)\right|_{t=1/x}.$$

Suppose first that $0 < c < d$. By Lemma 5.1, $J_{c,d}$ is strictly concave. For $s = x + y$, strict concavity and continuity at zero imply

$$J_{c,d}(x) > \frac{y}{s} J_{c,d}(0) + \frac{x}{s} J_{c,d}(s)$$

and

$$J_{c,d}(y) > \frac{x}{s} J_{c,d}(0) + \frac{y}{s} J_{c,d}(s).$$

Adding these inequalities gives

$$J_{c,d}(x) + J_{c,d}(y) - J_{c,d}(x + y) > 1.$$

By taking the exponential proves (5.1).

If $c > d$, the function $J_{c,d}$ is strictly convex, so both inequalities reverse. This proves the reverse strict inequality.

If $c = d$, then

$$J_{d,d}(x) = 1 - x \log \Gamma(d),$$

and hence

$$J_{d,d}(x) + J_{d,d}(y) - J_{d,d}(x + y) = 1.$$

Thus the quotient equals e.

Finally, taking $x = y = t$ and letting $t \downarrow 0$ gives

$$\log \left(\frac{D_{c,d}(1/t)^{2t}}{D_{c,d}(1/(2t))^{2t}} \right) = 2J_{c,d}(t) - J_{c,d}(2t) \longrightarrow 1.$$

Therefore the quotient tends to e. The lower bound cannot be increased when $0 < c < d$, and the upper bound cannot be decreased when $c > d$. □

For $d = 1$, Theorem 5.2 reduces to the Alzer–Kwong inequality ([2, Theorem 4.3]). In higher dimensions, the critical parameter is $c = d$, at which the diagonal series becomes a constant multiple of the exponential function.

Data availability

No data was used for the research described in the article.

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